

Requirements — Bridge_MT4_File.mq4

Overview

Bridge_MT4_File.mq4 is a standalone Expert Advisor for MetaTrader 4 that turns MT4 into a generic file-driven order executor. It does not depend on Python or any external runtime — it can be tested entirely with manually created text files.

User Stories

1. Periodic File Polling

Given the EA is attached to a chart and initialized,
When the configured poll interval elapses (default: 1 second),
Then the EA scans the `BridgeFolder` directory for `*.txt` action files.

2. Action File Parsing

Given a `*.txt` file exists in `BridgeFolder`,
When the EA picks it up during a poll cycle,
Then the EA reads it as a `key=value` text file, extracting: `id`, `asset`, `action`, `side`, `size`, `order_type`, `sl`, `tp`, `magic_number`, `valid_until`, `comment`. Unknown keys are silently ignored.

3. Required Field Validation

Given an action file has been parsed,
When any of the required fields (`id`, `action`, `asset`) are missing or empty,
Then the EA writes a `REJECTED` feedback file with `message=MissingRequiredField` and archives/deletes the action file without executing any trade.

4. Symbol Filtering

Given `OnlyCurrentSymbol = true` is configured,
When an action file specifies an `asset` that does not match the chart's symbol,
Then the EA rejects the action with `message=SymbolMismatch`.

Given `OnlyCurrentSymbol = false` (default),
When any valid symbol is specified in the action file,
Then the EA uses that symbol for execution.

5. Lot Size Validation

Given an `OPEN` action with `size` specified,
When `size > MaxLotsPerTrade`,
Then the EA rejects the action with `message=LotSizeExceeded`.

6. Spread Validation

Given an `OPEN` action has passed lot size validation,
When the current spread in points exceeds `MaxSpread`,
Then the EA rejects the action with `message=MaxSpreadExceeded`.

7. Time Validity Check

Given an action file contains a `valid_until` field (ISO 8601 format),
When the current server time is past `valid_until`,
Then the EA rejects the action with `message=ActionExpired`.

Given `valid_until` is blank,
When the action is processed,
Then no time validation is applied — the action is always considered valid.

8. Confirmation Prompt (Optional)

Given `AskForConfirmation = true` is configured,
When an action passes all validation,

Then the EA shows a `MessageBox` dialog asking the trader to confirm before sending the order.

If the trader clicks Cancel, the EA writes `REJECTED` with `message=UserCancelled` and archives the file.

9. OPEN — Market Order Execution

Given `action=OPEN`, `order_type=MARKET`, and all validations pass,

When the EA executes,

Then:

- It calls `RefreshRates()` before sending.
 - For `side=BUY`: `OrderSend` with `OP_BUY` at `Ask`.
 - For `side=SELL`: `OrderSend` with `OP_SELL` at `Bid`.
 - `sl` and `tp` are passed if non-zero; otherwise `0.0`.
 - Magic number: use `magic_number` from file if > 0 , else fall back to `MagicNumberBase`.
 - On success (ticket > 0): write `FILLED` feedback with ticket number and fill price.
 - On failure: capture `GetLastError()`, write `REJECTED` feedback with MT4 error code.
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10. CLOSE_ALL Action

Given `action=CLOSE_ALL` and all validations pass,

When the EA executes,

Then:

- Iterate open orders from `OrdersTotal()-1` down to `0`.
 - Filter by `asset` symbol (respecting `OnlyCurrentSymbol`).
 - Optionally filter by `magic_number` if provided in the file.
 - Close `OP_BUY` orders at `Bid`, `OP_SELL` orders at `Ask`.
 - Track count of successful and failed closes.
 - Write `FILLED` feedback summarising closed tickets.
 - If no matching orders found: write `REJECTED` with `message=NoOrdersToClose`.
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11. Feedback File Writing

Given any action is processed (success or failure),

When the outcome is determined,

Then the EA writes a feedback file to `FeedbackFolder` named `{id}_result.txt`

containing: `id`, `status`, `asset`, `action`, `side`, `size`, `tickets`, `avg_price`, `message`, `error_code`.

12. Action File Archiving

Given an action file has been processed (any outcome),

When feedback has been written,

Then the EA moves the action file to `ArchiveFolder` using `FileMove`. If `FileMove` fails, it falls back to `FileDelete` to prevent reprocessing.

13. Initialization Logging

Given the EA initialises on a chart,

When `OnInit` runs,

Then the EA logs the configured `BridgeFolder`, `FeedbackFolder`, and `ArchiveFolder` paths to the Experts tab. It sets the poll timer via `EventSetTimer(PollIntervalSeconds)`.

14. Clean Shutdown

Given the EA is removed from a chart or MT4 is closed,

When `OnDeinit` runs,

Then the EA calls `EventKillTimer()` and logs a shutdown message. No open file handles are left dangling.

15. Verbose Logging

Given `VerboseLogging = true`,

When any significant step occurs (file found, parsed, validated, executed, feedback written, archived),

Then the EA prints a descriptive message to the MT4 Experts log.

Out of Scope (v1)

- Limit or stop orders (`order_type` other than `MARKET`)

- Partial closes
- Trailing stop management
- Python runtime dependency
- Network connectivity
- MT5 support (separate EA)